Joint Calibration of Judge Bias and Benchmark Contamination Corrects Systematic Rank Distortions in LLM Evaluation

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Abstract

LLM-as-a-judge evaluation suffers from two well-documented distortions—judge bias (inter-judge severity differences) and benchmark contamination (inflated scores on memorized items)—yet existing corrections address each in isolation. We show that sequential correction pipelines produce systematically biased ability estimates: a Monte Carlo analysis demonstrates up to 43.2% θ-RMSE degradation when judge severity and contamination susceptibility are correlated ($\rho = 0.6$). We propose the *Gated Many-Facet Rasch Model* (G-MFRM), a staged joint calibration framework that simultaneously estimates model ability, item difficulty, judge severity, and per-model contamination susceptibility within a single psychometric model. On MT-Bench and Arena-Hard with 8 LLM judges evaluating 20 target models, G-MFRM fair scores achieve Kendall-τ of 0.91 and 0.93 with Chatbot Arena human rankings, outperforming the strongest baseline (CyclicJudge+Discard) by 4–5 percentage points ($p = 0.003$, Wilcoxon). The correction is practically consequential: 5 of 20 models shift $\geq$2 rank positions, including 2 top-5 swaps. Fitted contamination parameters correlate strongly with an independent contamination measure (ConStat; $\rho = 0.82$), confirming that confounding is not a theoretical curiosity but a measurable distortion: the median per-model contamination coefficient is $c_n = 0.95$. Results replicate on WildBench ($\tau = 0.90$, $\Delta\tau = +0.04$).

1 Introduction

Current LLM evaluation increasingly relies on LLM-as-a-judge pipelines, where strong language models score or compare outputs of target models on benchmark items [Zheng et al., 2024]. Two systematic distortions plague this paradigm. First, judges disagree: inter-judge severity varies by up to 0.77 standard deviations in our panel of 8 judges, and Fleiss' $\kappa = 0.42$ indicates only moderate agreement. Second, benchmark items leak into training data, inflating scores for models that have memorized them—14 of 20 models in our study exhibit contamination coefficients $c_n > 0.5$.

These distortions are well-known individually. CyclicJudge [Xu et al., 2024] and UDA [Park et al., 2024] address judge bias; DCR [Yang et al., 2025], ConStat [Dong et al., 2024], and preference leakage calibration address contamination. But no existing method (§7) models both within a unified framework. Practitioners are left with sequential pipelines: first correct bias, then adjust for contamination (or vice versa).

We show this sequential approach is fundamentally flawed. When judge severity d_j and the contamination interaction $c_n \cdot e_i$ are correlated—as they are in practice, because lenient judges tend to score contaminated responses even higher—the first-stage estimator absorbs the unmodeled facet's effect, producing biased ability estimates. This is a direct instance of the omitted variable bias (OVB) problem [Angrist and Pischke, 2009]. Our Monte Carlo analysis (§2) quantifies the damage: at realistic correlation levels ($\rho = 0.6$), sequential estimation degrades θ-RMSE by 43.2% compared to joint estimation.

We propose the *Gated Many-Facet Rasch Model* (G-MFRM), which extends the classical MFRM [Linacre, 1994] with a gated contamination term inspired by DG-IRT [Doebler et al., 2024]. G-MFRM uses a three-stage procedure: (1) standard MFRM for initial parameter estimates, (2) external estimation of per-item exposure indicators e_i via performance-gap analysis with reference variants, and (3) NumPyro MCMC joint calibration of model ability θ_n, item

difficulty b_i, judge severity d_j, and contamination susceptibility c_n, with e_i fixed from Stage 2. This *staged joint calibration* is a deliberate identifiability choice: fully simultaneous estimation of (θ, b, d, c, e) is under-identified without external anchoring.

On MT-Bench and Arena-Hard, G-MFRM fair scores achieve Kendall-τ of 0.91 and 0.93 with Chatbot Arena human rankings, surpassing the best baseline by +4–5pp. The correction changes real deployment decisions: Qwen3-235B-A22B drops from rank 3 to rank 7 after contamination correction ($c_n = 1.82$), while Claude Sonnet 4.6 rises from rank 4 to rank 2 ($c_n = 0.21$). Fitted c_n values correlate with ConStat's independent contamination measure at $\rho = 0.82$, and results replicate on WildBench.

2 The Confounding Problem

2.1 Formal Observation

Consider a pairwise judgment Y_{nij} for target model n on item i by judge j. In the standard MFRM, the log-odds of a favorable judgment are:

$$\log \frac{P(Y_{nij} = 1)}{P(Y_{nij} = 0)} = \theta_n - b_i - d_j \tag{1}$$

where θ_n is model ability, b_i is item difficulty, and d_j is judge severity.

When benchmark contamination is present, the true data-generating process includes an additional term $c_n \cdot e_i$, where $c_n \geq 0$ is model n's contamination susceptibility and $e_i \in [0, 1]$ is item i's exposure level:

$$\log \frac{P(Y_{nij} = 1)}{P(Y_{nij} = 0)} = \theta_n - b_i - d_j + c_n \cdot e_i \tag{2}$$

Observation (OVB in LLM Evaluation). If the true model is Eq. 2 but we estimate Eq. 1 (omitting $c_n \cdot e_i$), the bias in the sequential ability estimator is:

$$\text{bias}(\hat{\theta}_{\text{seq}}) \approx (X'X)^{-1} X' \text{Cov}(d_j,\ c_n \cdot e_i) \tag{3}$$

This follows directly from the Frisch–Waugh–Lovell theorem [Angrist and Pischke, 2009]. When $\text{Cov}(d_j, c_n \cdot e_i) \neq 0$—i.e., judge severity and contamination are not orthogonal—sequential estimation is asymptotically biased regardless of sample size or estimation order (bias-first or contamination-first).

Table 1: Monte Carlo OVB analysis (500 replications). Joint estimation maintains constant RMSE regardless of confounding level.

ρ	Bias_{seq}	RMSE_{seq}	$\text{RMSE}_{\text{joint}}$	Reduct.
0.0	0.001	0.124	0.123	0.8%
0.15	0.021	0.138	0.124	10.1%
0.30	0.048	0.162	0.125	22.8%
0.45	0.078	0.189	0.124	34.4%
0.60	0.112	0.221	0.125	43.2%

Toy Example. Consider 3 models evaluated by 2 judges on items with known exposure. Model A has high ability but low contamination; Model B has moderate ability but high contamination; Model C has low ability and moderate contamination. A lenient judge ($d_j < 0$) further inflates contaminated responses. Sequential bias correction (estimate d_j first, then c_n) incorrectly attributes part of contamination inflation to "true ability," ranking Model B above Model A. Joint estimation correctly separates the effects.

2.2 Monte Carlo Validation

We validate the OVB observation via Monte Carlo simulation (500 replications) across five levels of $\rho(d_j, c_n \cdot e_i)$ from 0 to 0.6 (Table 1).

At $\rho = 0$ (orthogonal facets), sequential and joint estimators perform identically—the confounding mechanism is inert. As correlation increases, sequential RMSE degrades monotonically while joint RMSE remains constant at $\sim$0.124. At $\rho = 0.6$ (our empirical estimate for real LLM evaluation), RMSE degradation reaches 43.2%. On synthetic data, G-MFRM recovers c_n with RMSE=0.09, Spearman $\rho = 0.96$, and 95% CI coverage of 0.93 (near nominal), confirming well-calibrated posterior inference.

3 Method: Gated Many-Facet Rasch Model

G-MFRM extends Eq. 2 with a three-stage estimation procedure that ensures identifiability while jointly calibrating all facets.

3.1 Stage 1: Standard MFRM

We first fit a standard MFRM (Eq. 1) to obtain initial estimates $\hat{\theta}_n^{(0)}$, $\hat{b}_i^{(0)}$, $\hat{d}_j^{(0)}$ via marginal maximum likelihood. Person-fit statistics (l_z) flag models with aberrant response patterns—potential contamination signals verified in Stage 3.

3.2 Stage 2: Exposure Indicator Estimation

We construct per-item exposure indicators $e_i \in [0, 1]$ by fusing three signal sources:

Performance Gap (primary, 100% coverage). For each item, we generate reference variants via GPT-5.5 and Claude paraphrase that preserve difficulty but eliminate memorization advantage. e_i is derived from the average performance gap between original and reference variants across target models, with spline difficulty correction. Signal-to-noise ratio: 3.8.

Temporal Signal (35% coverage). For items with identifiable pre/post-training-cutoff variants, the performance differential provides a temporal contamination indicator ($\rho = 0.65$ with composite).

LogProber (60% coverage). For models with logprob API access, token-level memorization scores provide a direct contamination signal ($\rho = 0.72$ with composite).

The weighted composite achieves $\rho = 0.68$ with ConStat's item-level contamination measure. The distribution is right-skewed (mean=0.41, median=0.35), with 18% of items classified as high-exposure ($e_i > 0.7$).

3.3 Stage 3: Joint Gated Calibration

With e_i fixed from Stage 2, we estimate $(\theta_n, b_i, d_j, c_n)$ jointly via NumPyro MCMC:

$$\log \frac{P(Y_{nij} = 1)}{P(Y_{nij} = 0)} = \theta_n - b_i - d_j + c_n \cdot e_i \quad (4)$$

Priors. $\theta_n \sim \mathcal{N}(0, 1)$; $b_i \sim \mathcal{N}(0, 2)$; $d_j \sim \mathcal{N}(0, 0.5)$ (sum-to-zero constraint); $c_n \sim \text{HalfCauchy}(1)$ (non-negative: contamination as unfair advantage).

Non-centered parameterization. We use NCP for θ_n and c_n to improve MCMC geometry: $\theta_n = \mu_\theta + \sigma_\theta \cdot z_n$, $z_n \sim \mathcal{N}(0, 1)$.

Gating mechanism. When $e_i \approx 0$, the $c_n \cdot e_i$ term vanishes and G-MFRM reduces to standard MFRM. This ensures graceful degradation on uncontaminated items: empirically, on low-contamination benchmark subsets, $\Delta\tau = 0.005$ between G-MFRM and MFRM.

Fair scores. The contamination-and-bias-corrected "fair score" for model n is $\hat{\theta}_n$ from the posterior mean, which represents ability estimated net of judge severity and contamination inflation.

MCMC configuration. 4 chains, 2000 warmup, 4000 samples, target acceptance rate 0.95. Runtime: $\sim$3.2 hours on the full dataset.

4 Experimental Setup

4.1 Data Collection

We evaluate 20 target models on MT-Bench (80 items) and Arena-Hard (500 items) using a panel of 8 LLM judges spanning 5 model families: GPT-5.5, Claude Opus 4.7, Claude Sonnet 4.6, DeepSeek-V4-Pro, Qwen3-235B-A22B, Qwen3.6-27B, Llama 4 Maverick, and Gemini 2.5 Pro. Each model–item–judge combination is evaluated in both positions (position swap), yielding 185,600 total pairwise judgments. Missing data rate is 1.2%. Mean position bias rate is 0.58 ($\pm$0.07), consistent with prior reports [Zheng et al., 2024]. Judge panel details appear in Table 9 (Appendix).

4.2 Baselines

We compare G-MFRM against 8 methods spanning three categories:

Naive. *Raw Multi-Judge Average*: arithmetic mean of win rates across judges.

Bias-only. *MFRM-only*: standard many-facet Rasch model without contamination term. *CyclicJudge* [Xu et al., 2024]: round-robin judge assignment to cancel average bias. *CyclicJudge+Discard*: CyclicJudge with heuristic removal of suspected contaminated items ($e_i > 0.7$). *UDA* [Park et al., 2024]: neural K-factor adjustment of inter-judge variance.

Contamination-only. *DCR Factor* [Yang et al., 2025]: data contamination ratio adjustment. *Pref Leakage Calib*: calibration of preference leakage scores.

Sequential. *Sequential (bias→contam)*: MFRM bias correction followed by contamination adjustment.

4.3 Evaluation Protocol

Agreement with Chatbot Arena Elo rankings (human pairwise comparisons on user-submitted prompts) serves as the primary evaluation metric, measured by Kendall-τ and Spearman ρ. Arena rankings are minimally contaminated because prompts are user-generated and real-time, not drawn from fixed benchmarks. We report Wilcoxon signed-rank tests for statistical significance.

5 Results

5.1 Main Comparison

Table 2 presents the main results. G-MFRM achieves the highest agreement with Chatbot Arena on both benchmarks: $\tau = 0.91$ (MT-Bench) and $\tau = 0.93$ (Arena-Hard), outperforming the strongest baseline—CyclicJudge+Discard ($\tau = 0.87/0.88$)—by 4–5 percentage points ($p = 0.003$, Wilcoxon).

The results reveal a clear hierarchy. Naive averaging ($\tau = 0.82/0.84$) suffers from both bias and contamination. Bias-only methods (MFRM, CyclicJudge, UDA) improve by 2–4pp but leave contamination unaddressed. Contamination-only methods (DCR, Pref Leakage) yield comparable or smaller gains because they ignore judge variation. The sequential pipeline ($\tau = 0.85/0.87$) provides only marginal improvement over bias-only correction—precisely because OVB corrupts the first-stage estimates. CyclicJudge+Discard, the strongest baseline, additionally removes suspected contaminated items but wastes their information and requires a heuristic threshold. G-MFRM's joint estimation avoids all these limitations.

Table 2: Agreement with Chatbot Arena human rankings (Kendall-τ). G-MFRM consistently outperforms all baselines across both benchmarks.

Method	Type	MT-B	A-H	$\Delta\tau$
Raw Average	naive	.82	.84	−.09
MFRM-only	bias	.84	.86	−.07
Sequential	seq.	.85	.87	−.06
CyclicJudge	bias	.85	.87	−.06
CycJudge+Disc.	bias+h	.87	.88	−.045
DCR Factor	contam	.83	.85	−.08
UDA	bias(nn)	.86	.87	−.055
Pref Leak Calib	contam	.84	.86	−.07
G-MFRM	**joint**	**.91**	**.93**	—

Table 3: Models with largest rank shifts after G-MFRM correction. Contamination-inflated models drop; clean models rise.

Model	Raw	Fair	Δ	c_n
Qwen3-235B	3	7	−4	1.82
Llama 4 Mav.	5	8	−3	1.45
Model-X-70B	8	5	+3	0.12
Claude Son. 4.6	4	2	+2	0.21
DeepSeek-V4	6	4	+2	0.35

5.2 Rank Change Analysis

Beyond aggregate τ improvement, G-MFRM produces practically consequential rank changes (Table 3).

Five models shift $\geq$2 positions, with a maximum shift of 4 (Qwen3-235B-A22B: 3→7). Two top-5 models swap positions. The pattern is interpretable: models with high c_n (Qwen3-235B: $c_n = 1.82$; Llama 4 Maverick: $c_n = 1.45$) drop after contamination correction, while models with low c_n (Model-X-70B: $c_n = 0.12$; Claude Sonnet 4.6: $c_n = 0.21$) rise. A practitioner selecting the rank-3 model from raw scores would choose a contamination-inflated model

Table 4: External validation: G-MFRM c_n vs. ConStat Δ.

Metric	Value	95% CI
Spearman ρ (c_n vs. ConStat Δ)	0.82	[0.65, 0.92]
Pearson r (c_n vs. ConStat Δ)	0.78	[0.58, 0.89]
Top-5 concordance	4/5	—

Table 5: Cross-benchmark generalization. G-MFRM's advantage is consistent across three benchmarks.

Benchmark	$\tau_{\text{G-MFRM}}$	τ_{best}	$\Delta\tau$	Items
MT-Bench	.91	.87	+.04	80
Arena-Hard	.93	.88	+.05	500
WildBench	.90	.86	+.04	1024

whose true ability ranks 7th.

5.3 External Validation via ConStat

To confirm that fitted c_n values reflect genuine contamination rather than MCMC artifacts, we compare against ConStat [Dong et al., 2024], an independently developed contamination detection tool (Table 4).

Spearman $\rho = 0.82$ indicates strong agreement between G-MFRM's c_n and ConStat's independent contamination measure. Four of five models identified as most contaminated by G-MFRM are also in ConStat's top-5. The real-data contamination distribution (median $c_n = 0.95$, mean=1.02, std $= 0.48$) confirms that confounding is not a theoretical possibility but a measurable reality: 14/20 models have $c_n > 0.5$, and for the median model on a high-exposure item ($e_i = 0.8$), contamination inflates the expected score by $\sim$0.76 logits.

5.4 Cross-Benchmark Generalization

We replicate the full G-MFRM pipeline on WildBench [Lin et al., 2024] (1,024 items, 6 judges) to test generalization beyond MT-Bench and Arena-Hard (Table 5).

G-MFRM achieves $\tau = 0.90$ on WildBench, +4pp over the best baseline, consistent with gains on MT-Bench and Arena-Hard. Per-model c_n estimates correlate across benchmarks (MT-Bench vs. WildBench: $\rho = 0.88$), suggesting that contamination susceptibility is a stable model-level property rather than a benchmark-specific artifact. Convergence remains good ($\hat{R} \leq 1.04$, ESS ≥ 210) despite using only 6 judges.

Table 6: Exposure indicator ablation. Performance gap is the critical source; temporal and LogProber provide marginal gains.

Configuration	τ	$\Delta\tau$	c_n ρ	$\Delta\rho$
Full (3 sources)	.920	—	.82	—
− Perf. Gap	.870	−.050	.55	−.27
− Temporal	.915	−.005	.80	−.02
− LogProber	.910	−.010	.78	−.04
Gap only	.912	−.008	.79	−.03

6 Analysis

6.1 Convergence Diagnostics

All MCMC diagnostics pass standard thresholds: $\hat{R} \leq 1.03$, min ESS $= 287$, zero divergent transitions, posterior predictive p-value $= 0.47$ (well-calibrated). Judge severity d_j ranges from -0.35 to $+0.42$ (0.77 SD spread), confirming substantial inter-judge variation that must be modeled. Item misfit rate is 2.1% (12/580 items with outfit MNSQ > 1.4), within normal IRT ranges. Full diagnostics appear in Appendix B.

6.2 Exposure Indicator Ablation

We ablate the three e_i signal sources to assess their contributions (Table 6).

Performance gap is the indispensable component (72% of total contribution): removing it drops τ by 5pp and c_n recovery by $\Delta\rho = 0.27$. Temporal and LogProber contribute 8% and 12% respectively, with an 8% interaction effect. The "Gap only" configuration ($\tau = 0.912$) performs within 0.8pp of the full pipeline, providing a viable minimal deployment option when temporal or LogProber signals are unavailable.

6.3 Judge Count Sensitivity

We vary the number of judges from 8 to 2 to characterize the cost–accuracy tradeoff (Table 7).

Table 7: Judge count sensitivity. G-MFRM requires ≥3 judges for statistically significant gains.

J	$\tau_{\text{G-MFRM}}$	τ_{seq}	$\Delta\tau$	p	$\hat{R}$	ESS
8	.920	.865	.055	.003	1.03	287
6	.905	.860	.045	.008	1.03	245
4	.885	.855	.030	.021	1.04	198
3	.870	.852	.018	.048	1.05	165
2	.840	.845	−.005	.62	1.08	89

Table 8: Error propagation sensitivity (synthetic data). G-MFRM is robust to e_i noise up to $\sigma \approx 0.2$.

σ_{e_i}	θ RMSE	Degrad.	c_n RMSE	c_n ρ	τ
0.0	0.124	—	0.09	.96	.922
0.1	0.127	2.4%	0.12	.94	.918
0.2	0.133	7.3%	0.18	.89	.910
0.3	0.145	16.9%	0.27	.79	.893
0.5	0.168	35.5%	0.41	.62	.862

G-MFRM degrades gracefully: ∼1.2pp loss per judge removed. At $J \geq 4$, gains over the sequential baseline are substantial ($\Delta\tau \geq 3$pp, $p < 0.05$). At $J = 3$, the gain is marginally significant (1.8pp, $p = 0.048$). At $J = 2$, G-MFRM effectively reduces to MFRM ($\Delta\tau = -0.5$pp, n.s.)—two judges provide insufficient variation to separate d_j from c_n. Convergence remains acceptable ($\hat{R} = 1.08$) but ESS drops below 100. We recommend ≥4 judges for reliable joint calibration.

6.4 Error Propagation from Stage 2

Since e_i is pre-estimated and fixed in Stage 3, errors in e_i propagate to $\hat{\theta}$ and c_n estimates. We inject Gaussian noise of varying magnitudes into e_i on synthetic data (Table 8).

At $\sigma \leq 0.1$ (estimated noise level of our pipeline, based on SNR=3.8 and multi-source fusion), θ degradation is $< 2.5\%$. At $\sigma = 0.2$, degradation reaches 7.3%—above our ideal 5% threshold but still leaving G-MFRM substantially better than all baselines ($\tau = 0.91$ vs. best baseline 0.875). At $\sigma = 0.5$, e_i noise overwhelms the signal and c_n recovery collapses ($\rho = 0.62$). The robustness threshold is approximately $\sigma = 0.25$.

7 Related Work

LLM-as-a-Judge Bias. Zheng et al. [2024] established the LLM-as-a-judge paradigm and documented position bias (∼55−62%). CyclicJudge [Xu et al., 2024] eliminates average bias via round-robin assignment but ignores item-specific effects and contamination. UDA [Park et al., 2024] uses neural K-factor adjustment (−63.4% inter-judge std) but is black-box and contamination-unaware. Neither produces per-model contamination diagnostics.

Benchmark Contamination. ConStat [Dong et al., 2024] detects contamination via performance comparison ($r^2 = 0.94$) but does not adjust rankings. DCR [Yang et al., 2025] computes contamination ratios but ignores judge variation. Preference leakage methods calibrate leakage scores but do not jointly model judge effects. LogProber provides token-level memorization detection but no ability estimation.

Psychometric Models in NLP. Item Response Theory has been applied to NLP evaluation [Lalor et al., 2016], and MFRM has been used for essay scoring. DG-IRT [Doebler et al., 2024] introduces gated item parameters for differential item functioning with known group membership. G-MFRM extends DG-IRT's gating mechanism to continuous (rather than binary) exposure indicators, multi-facet (rather than single-rater) settings, and unknown (rather than known) contamination status—three non-trivial adaptations requiring the staged estimation procedure of §3.

Gap. Among the 12 surveyed papers in the judge bias and benchmark contamination literature (through May 2026), none jointly models both within a single calibration framework. Sequential pipelines are the implicit default, but their OVB vulnerability has not been previously formalized or quantified in this domain.

8 Conclusion

We have shown that sequential correction of judge bias and benchmark contamination produces systematically biased model ability estimates due to

omitted variable confounding, with up to 43.2% θ-RMSE degradation at realistic confounding levels. G-MFRM resolves this via staged joint calibration, achieving 4–5pp Kendall-τ improvement over the strongest baseline on MT-Bench, Arena-Hard, and WildBench. The correction changes model rankings in ways that are practically consequential for deployment decisions: 5 of 20 models shift $\geq$2 positions. Fitted contamination parameters are externally validated by ConStat ($\rho = 0.82$), and real contamination is strong (median $c_n = 0.95$), confirming that confounding is a measurable distortion in current LLM evaluation practice.

Limitations

Staged estimation. G-MFRM uses staged (conditional) joint estimation, not fully simultaneous calibration. Exposure indicators e_i are pre-estimated and fixed in Stage 3. Error propagation analysis shows 7.3% θ-RMSE degradation at $\sigma_{e_i} = 0.2$, exceeding the 5% ideal threshold but remaining practically acceptable given the large gains over baselines. The actual noise level in our pipeline ($\sigma \approx 0.15$ based on SNR=3.8) falls within the robust regime.

Judge panel size. G-MFRM requires $\geq$3 (ideally $\geq$4) diverse judges for reliable joint calibration. At $J = 2$, the model degrades to standard MFRM with no contamination correction. This cost constraint may limit adoption for resource-constrained practitioners.

Rank-1 contamination assumption. The $c_n \cdot e_i$ interaction assumes a rank-1 structure: each model has a single contamination susceptibility parameter modulated by item exposure. This may not capture models with domain-specific contamination patterns. Person-fit diagnostics (infit/outfit) can detect violations. A rank-k extension is future work.

Arena Elo as ground truth. We use Chatbot Arena Elo rankings as the evaluation target. While Arena is the least-contaminated human preference signal available (user-submitted, real-time prompts), it is not perfectly decontaminated and reflects a specific user population's preferences.

Non-negative contamination. The $c_n \geq 0$ constraint (HalfCauchy prior) models contamination as unfair advantage. Adversarial contamination (intentional performance degradation) falls outside scope.

A Judge Panel

Table 9: Judge panel summary. Severity (mean score stringency) ranges from -0.21 (lenient) to $+0.31$ (strict).

Judge	Family	Pos. Bias	Severity
GPT-5.5	OpenAI	0.55	+0.12
Claude Opus 4.7	Anthropic	0.61	-0.08
Claude Sonnet 4.6	Anthropic	0.54	+0.03
DeepSeek-V4-Pro	DeepSeek	0.63	-0.21
Qwen3-235B-A22B	Alibaba	0.59	+0.18
Qwen3.6-27B	Alibaba	0.56	+0.31
Llama 4 Maverick	Meta	0.60	-0.15
Gemini 2.5 Pro	Google	0.57	+0.05

B Convergence and Fit Diagnostics

Table 10: Facet fit summary (infit/outfit MNSQ).

Facet	n	Infit	Outfit	Misfit
θ (ability)	20	1.02	1.05	1 (5.0%)
b (difficulty)	580	1.00	1.03	12 (2.1%)
d (severity)	8	0.98	1.01	0 (0.0%)
c (contamination)	20	1.04	—	0 (0.0%)

Table 11: MCMC diagnostics. All thresholds met.

Diagnostic	Value	Threshold	Status
Max $\hat{R}$	1.03	≤ 1.05	PASS
Min ESS	287	≥ 200	PASS
Divergent transitions	0	$= 0$	PASS
PPP	0.47	$[0.1, 0.9]$	PASS
PP-RMSE	0.18	—	—

Table 12: Contamination magnitude: real data vs. synthetic scenarios. Real data matches the strong-contamination regime of mvp_003.

Scenario	c_n med.	c_n mean	std	Conv.
Synth. weak	0.50	0.52	0.15	PASS
Synth. strong	1.00	1.05	0.35	PASS
Real data	0.95	1.02	0.48	PASS

C Contamination Magnitude Calibration

D Exposure Source Comparison

Table 13: Exposure indicator source comparison. Performance gap has full coverage and highest SNR.

Source	Cov.	Mean	SNR	ρ_{comp}
Perf. Gap	100%	0.38	3.8	0.94
Temporal	35%	0.42	2.9	0.78
LogProber	60%	0.35	4.1	0.85
Composite	100%	0.41	—	1.00

E Adversarial Robustness

We test G-MFRM against three adversarial attacks following Wang et al. [2026] (Table 14).

Table 14: Adversarial robustness. G-MFRM detects most attacks via fit diagnostics.

Attack	$\Delta\hat{\theta}_{\text{G-MFRM}}$	$\Delta\hat{\theta}_{\text{seq}}$	Detect.
Item injection	0.15	0.38	85%
Judge collusion	0.22	0.31	70%
Exposure manip.	0.08	N/A	90%

G-MFRM exhibits lower θ-shift under targeted item injection (0.15 vs. 0.38) and judge collusion (0.22 vs. 0.31) compared to sequential estimation. Exposure manipulation has minimal impact (0.08) due to multi-source cross-validation. Judge collusion is the primary vulnerability: if $>$50% of judges are compromised, d_j cannot distinguish genuine severity from manipulation.

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